

Sheet No. 1

NATIONAL INSTITUTE OF TECHNOLOGY, KURUKSHETRA
THEORY EXAMINATION
 Question Paper

Month and Year of Examination : **JULY 2026**

Programme : **B.Tech**

Subject: **Analog Electronics**

Course No: **RAPC-209**

Number of Questions to be Attempted: **5**

Total No. of Questions: **5**

Semester: **4th**

Maximum Marks: **50**

Time Allowed: **3 Hrs.**

Total No. of Pages Used: **3**

Unless stated otherwise, the Symbols have their usual meanings in context with the Subject. Assume suitably and state, additional data required, if any.

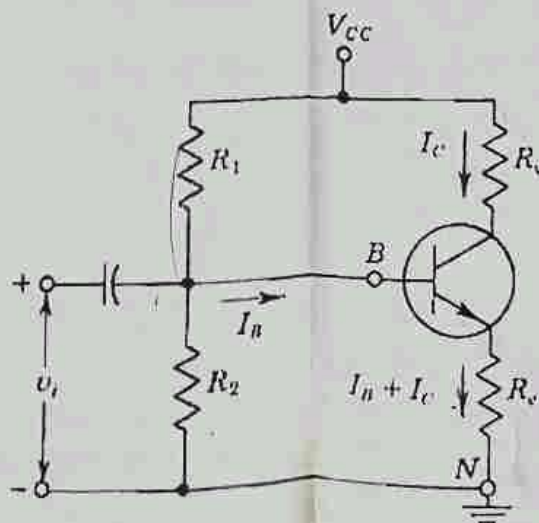
The Candidates, before starting to write the solutions, should please check the Question Paper for any discrepancy, and also ensure that they have been delivered the question paper of **right course no.** and **right subject title.**

Note: Attempt all the questions with equal marks.

- 1a. Critically discuss the classification of amplifiers on the basis of magnitudes of input and output impedances relative to the source and load impedances, respectively alongwith the suitable illustrations and mathematics.

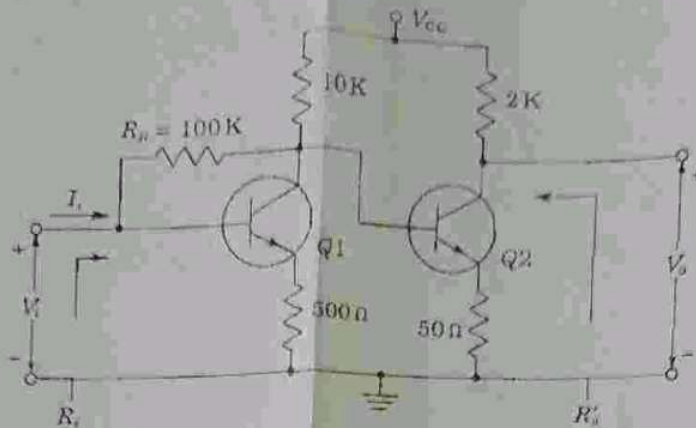
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- 1b. Explain biasing and its need in a transistor amplifier.
 A PNP germanium transistor is used in the self-biasing arrangement similar to that shown in the Fig. below. The circuit component values are $V_{cc} = 4.5 \text{ V}$, $R_c = 1.5 \text{ K}$, $R_e = 0.27 \text{ K}$, $R_2 = 2.7 \text{ K}$ and $R_1 = 27 \text{ K}$. If $\beta = 44$
 i) find the quiescent point ii) recalculate these values if the base spreading resistance of 690Ω is taken into account.



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- 2a. Explain the need of the small signal model of a BJT. Also, discuss why only h-parameters are used for the purpose.
 Find A_i , A_v , R_i and R_o for the two-stage cascade shown in the Fig. below.

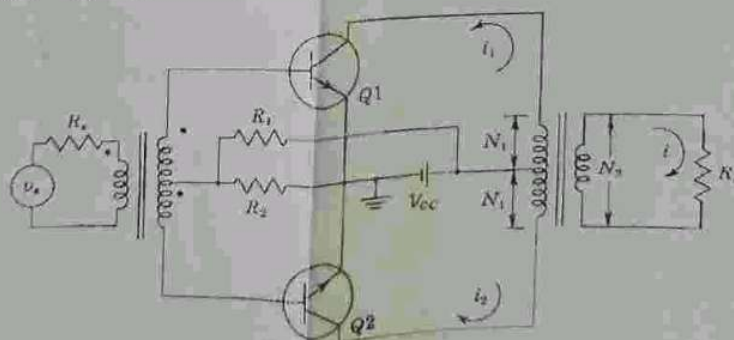


- 2b. Discuss the factors that come up when a BJT is modeled at high frequency as compared to its small signal h-parameter model with the help of an example of high frequency model of a CE stage.

Given a Ge PNP transistor whose base width is 10^{-4} cm. At room temperature and for a dc emitter current of 2 mA, find i) the emitter diffusion capacitance ii) f_T

- 3a. Define distortion in amplifiers. Also discuss the causes of various types of distortion with the help of suitable illustrations and mathematics.

The idealized push-pull class B power amplifier shown in the Fig. below has $R_2 = 0$, $V_{CC} = 20$ V, $N_2 = 2N_1$ and $R_L = 20 \Omega$, and the transistors have $h_{FE} = 20$. The input is a sinusoid. For the maximum output signal at $V_m = V_{CC}$, determine i) the output signal power ii) the collector dissipation in each transistor.



- 3b. Explain why a differential amplifier has been chosen to be a basic building block of an OPAMP IC. Also, define its various performance parameters along with mathematical expressions and significance of each. If a waveform has a positive peak of magnitude V_1 and a negative peak of magnitude V_2 , draw a circuit using two peak detectors whose output is equal to the peak-to-peak value $V_1 - V_2$

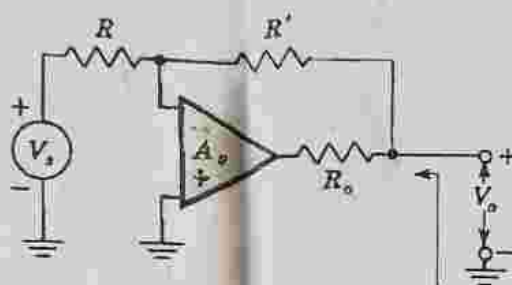
4. Define feedback in amplifiers and discuss in detail relative advantages and disadvantages of negative and positive feedback with suitable illustrations and examples.
An amplifier with open loop voltage gain $A_v = 1000 \pm 100$ is available. It is necessary to have an amplifier whose voltage gain varies by no more than $\pm 0.1\%$ i) find the reverse transmission factor β of the feedback network used ii) find the gain with feedback.

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OR

Enumerate different linear and nonlinear applications of OPAMP and discuss any two of them, one from each type, in detail with suitable illustrations and mathematics.

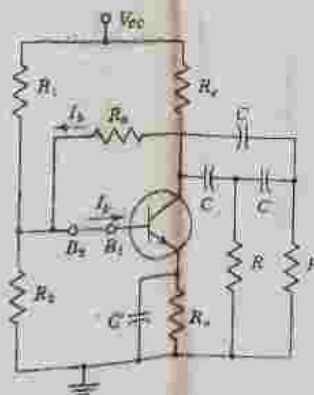
Show that $V_o = \frac{1}{R_o} \left(1 - A_v \frac{R}{R+R'} \right) + \frac{1}{R+R'}$ for the circuit shown in the Fig. below



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- 5a. Compare sinusoidal and non-sinusoidal oscillators. Discuss each case in detail with the help of suitable examples and their practical significance.

Show that the frequency of oscillation is $f = \frac{1}{2\pi RC} \frac{1}{\sqrt{6+4k}}$; $k = \frac{R_c}{R}$ in the circuit shown in the Fig. below. Also, find $(h_{fe})_{min}$ for the transistor used.



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- 5b. Compare active and passive filters critically with the help of illustrations and examples. Also discuss in detail design guidelines in each case.

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